HYDRO NAVIX

MARINE & OFFSHORE ENGINEERING

COMPANY PROFILE

Hydronavix — Engineering Excellence Across the Deep.

we craft smart, reliable marine and offshore engineering solutions built for the real challenges of the sea. From concept to completion, we bring precision, safety, and innovation to every wave serving maritime industries worldwide with engineering that lasts.

Modern Marine Engineering

Beyond the horizon lies more than water. It's where our engineering comes alive. At Hydronavix, we turn complexity into clarity, crafting offshore solutions that are as refined as they are resilient.

Infinite Innovation. Offshore Defined.

Echo-Class Engineering Systems

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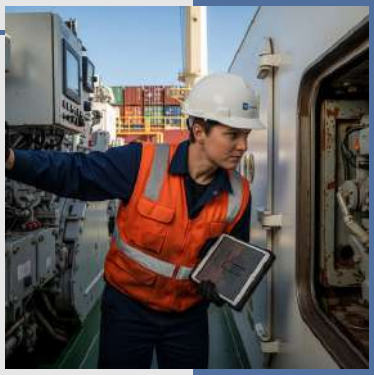

OUR VISION

To shape a globally adaptive marine future by pioneering intelligent infrastructure, advancing environmental resilience, and building long-term value through transformative engineering

OUR MISSION

To execute high-impact technical programs through data-rich modeling, modular project strategies, immersive spatial visualization, and next-gen automation enabling faster decision-making, enhanced uptime, and lifecycle-driven asset performance across maritime domains.

Primary Scope

01 Heavy Lift & Load-Out Engineering

- Structural weighing and weight control studies
- Rigging analysis, lifting plans, and crane capacity verification
- Heavy-lift engineering for modules, jackets, topsides, and large steel structures
- Load-out engineering (SPMT, skid, and crane-based operations)
- Float-in / float-out engineering and barge stability assessments
- Seafastening design and transport engineering support
- Coordination with class, third-party authorities, and client teams
- Lift execution methodology, risk assessments, and contingency planning
- On-call engineering support during critical lifting and load-out operations

Additional Scope

02 ENGINEERING & DESIGN

- 2D/3D Marine Design & Modeling
- Detailed Engineering
- Naval Architecture & Marine Simulation
- Ballasting & Mooring Plans
- As-Built Drafting
- Marine Equipment Integration Studies

03 DIGITAL SOLUTIONS

- AI-Based 2d to 3D Model Conversion (from GA Drawings)
- Marine Digital Twin Development
- VR/AR Visualization & Interactive Walkthroughs
- 3D Laser Scanning & Point Cloud Processing
- Remote Monitoring & Predictive Maintenance
- Smart Ship System Development
- Simulation-Driven Design Validation

04 FIELD & INSPECTION SERVICES

- Drone-Based Aerial, Underwater & Thermal Inspections
- Certified QA/QC Inspection
- Dimensional Control & Survey
- Engineering Audits & Site Reports
- Third-Party Verification Support
- Consultancy services for inline inspection tool design

05 TECHNICAL SUPPORT & PROJECT SUPPORT

- Certified Engineers & Draughtsmen (Marine/Offshore)
- Short-Term & Long-Term Project deployment
- Offshore Deployment (Yard & Vessel)
- Freelance & Site-Based Resources

06 Marine Automation Design, Engineering, Supply, Installation & Commissioning

Product and Service Portfolio:

1.MCC Panels – Design, Engineering & Supply
- DOL Starter
- Star-Delta Starter
- Soft Starter

(Panels engineered in-house and supplied through approved manufacturing partners)
2.PLC & Automation Control Panels - Engineering & System Integration
3.VFD & Drive Panels – Design, Configuration & Supply
4.SCADA / HMI Consoles & Operator Panels
5.Marine & Offshore Control Panels (Machinery control, alarm & monitoring systems)
6.Servo & Stepper Motion Control Panels (Design, programming & system integration)
7.Field Wiring, Installation & On-Site Integration
8.PLC/SCADA & Complete Automation Programming

Testing & Commissioning

- ***Factory Acceptance Test (FAT):***
  Complete witnessable testing of control logic, interlocks, I/O checks and power circuits, performed at approved partner facilities or client-designated locations prior to dispatch.

- ***Site Acceptance Test (SAT) / Commissioning:***
  On-site installation support, loop checks, system integration with PLC / SCADA,
  functional testing and performance validation at project location.

- ***Documentation:*** O&M manuals, as-built drawings, I/O lists, spare parts list,
  and PLC / SCADA program backup provided as per project scope or client request.

Note: Panel fabrication and factory testing are executed through HydroNavix-approved manufacturing partners in compliance with project, class and statutory requirements.

07 SHIPBUILDING & FABRICATION (INDIA OPERATIONS)

- Primary Scope – Executed through Indian Entity & Approved Fabrication Partners
- Hull steel fabrication for newbuild vessels and marine structures
- Block fabrication and modular ship construction methodology
- Structural steel fabrication for marine and offshore assets
- Plate cutting, rolling, forming and CNC profiling
- Sub-assemblies, unit fabrication, major block assembly and erection support
- Deck structures, bulkheads, stiffeners, girders and foundations
- Fabrication of barges, pontoons, tugboats and workboats
- Pile fabrication, pile cluster fabrication and jacket foundation components

- Fabrication of tubular piles, pile sleeves, stabbing guides and pile caps
- Welding execution as per approved WPS
- (SMAW, FCAW, SAW and approved special processes)
- Dimensional control, fit-up inspection and alignment verification
- NDT coordination including UT, MT, PT and RT
- QA/QC implementation in compliance with IACS Class requirements
- Topside structural fabrication for offshore platforms and modules
- Afloat repair, modification and structural strengthening works
- On-site and jetty-side fabrication support during afloat operations
- Class-ready fabrication with full documentation and material traceability

INDUSTRIES WE SERVE

- Shipbuilding & Ship Repair
- Offshore Oil & Gas
- Port Infrastructure & Dredging
- Renewable Offshore Wind & Energy
- Marine EPC & Fabrication Yards
- Subsea & Underwater Assets
- Digital Modernization of Legacy Ships

TOOLS & TECHNOLOGIES

HydroNavix Marine continues to expand its digital ecosystem by investing in emerging technologies to stay ahead of global marine engineering demands.
Innovation is not optional — it's engineered into everything we do.

- ***CAD Tools:*** ***(For 2D/3D drafting, detailing, and ship/offshore module design)***
  AutoCAD (Autodesk),AutoCAD Plant 3D (Autodesk), AVEVA (Marine / PDMS), CADMATIC, Tekla Structures, Ship Constructor, Maxsurf, Autodesk Inventor, Strand7, Pipe Flow Expert, Rhinoceros (Rhino), 3ds Max (Autodesk), AutoHydro

- ***Hardware:*** *Faro/Leica 3D Scanners, DJI Drones*

- ***Naval Architecture :*** *Autoship, GHS, Maxsurf*

- ***Simulation & Analysis Tools:*** ***(For hydrodynamics, structural integrity, mooring, pipe flow, and offshore behavior)***
  ANSYS, SACS (Offshore structural analysis), STAAD.Pro (Structural and load analysis), AutoHydro (Stability analysis), Maxsurf Stability / Motions (Hydrostatics and hydrodynamics), OrcaFlex (Mooring, riser & cable simulation), MOSES (Marine operations & floatover simulation), Optimoor (Mooring layout & analysis), Pipe Flow Expert, TEKLA (Simulation-integrated modeling), AVEVA Offshore / E3D, CADMATIC (Outfitting Simulation), Rhinoceros + Orca3D Plugin

- ***Software:*** *Unity, Unreal Engine for AR/VR*

From data to decisions — faster

We integrates smart analytics with engineering workflows, enabling real-time insights that drive precision offshore.

HYDRONAVIX MARINE AND OFFSHORE ENGINEERING - F.Z.C

We see the ocean not as a boundary, but as a frontier of intelligent engineering. At Hydronavix, we harness the power of automation, AI, and immersive technology to build offshore systems that think ahead. From real-time diagnostics to digitally enhanced training environments, our solutions are crafted for resilience, responsiveness, and long-term performance. The future isn't somewhere we wait for we engineer it.

READY TO TRANSFORM YOUR OFFSHORE OPERATIONS?

CONTACT US TO DISCUSS YOUR NEXT PROJECT.

CONTACT US

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LET'S START ENGINEERING YOUR MARINE & OFFSHORE ADVANTAGE TODAY.

Whether you're launching a new project, upgrading critical systems, or expanding offshore operations — Hydronavix is your trusted engineering partner, every step of the way.